

Foundation Language Models vs. Fine-Tuned Language Models

Foundation language models, such as the aforementioned MT-NLG and GPT-3, are what is usually referred to when discussing LLMs. They're trained on vast amounts of data and can perform a wide variety of NLP tasks, from answering questions and generating book summaries to completing and translating sentences.

Thanks to their size, foundation models can perform well even when they have little domain-specific data at their disposal. They have good general performance across tasks but may not excel at performing any one specific task.

Fine-tuned language models, on the other hand, are large language models derived from foundation LLMs. They're customized for specific use cases or domains and, thus, become better at performing more specialized tasks.

Apart from the fact that fine-tuned models can perform specific tasks better than foundation models, their biggest strength is that they are lighter and, generally, easier to train. But how does one actually fine-tune a foundation model for specific objectives?

Currently, the most popular method is customizing a model using parameter-efficient customization techniques, such as p-tuning, prompt tuning, adapters, and so on. Customization is far less time-consuming and expensive than fine-tuning the entire model, although it may lead to somewhat poorer performance than other methods. Customization methods are further discussed in [Part 3](#).

Evolution of Large Language Models

AI systems were historically about processing and analyzing data, not generating it. They were more oriented toward perceiving and understanding the world around us rather than on generating new information. This distinction marks the main difference between *Perceptive AI* and *Generative AI*, with the latter becoming increasingly prevalent since around 2020, or after companies started adopting transformer models and developing increasingly more robust LLMs at a large scale.

The advent of large language models further fueled a revolutionary paradigm shift in the way NLP models are designed, trained, and used. To truly understand this, it may be helpful to compare large language models to previous NLP models and how they worked. For this purpose, let's briefly explore three regimes in the history of NLP: pre-transformers NLP, transformers NLP, and LLM NLP.

1. **Pre-transformers NLP** was mainly marked by models that relied on human-crafted rules rather than machine learning algorithms to perform NLP tasks. This made them suitable for simpler tasks that didn't require too many rules, like text classification, but unsuitable for more complex tasks, such as machine translation. Rule-based models also performed poorly in edge-case scenarios because they couldn't make accurate predictions or classifications for never-before-seen data for which no clear rules were set. This problem was somewhat solved with simple neural networks, such as RNNs and LSTMs, developed during the later phases of this period. RNNs and LSTMs could memorize past data to a certain extent and, thus, provide context-dependent predictions and